In-Car Voice Assistant

Ablation and robustness report

Target speaker extraction -> presence -> wake word -> intent

Study date	Evidence status	Training
2026-10-10 update	**Proxy diagnostics**	**Bounded candidates complete**

Executive finding

The bounded dual-domain Stage 1 and Stage 2 candidates improve the AISHELL-5 and CAVEMOVE extraction regression and require a recalibrated presence threshold. On the current eight-scene cross-corpus proxy, the calibrated full path reaches 100% wake recall, 12.5% intent accuracy, and 0 false actions/hour. A raw post-wake intent handoff raises proxy intent accuracy to 37.5% with 0 false actions/hour in the standard passenger condition, but not in the harsher passenger condition. A 0.22 speaker gate plus 0.53 intent-confidence gate removes that residual action in a partial-speaker diagnostic, but is not calibrated for production. These are not real-driver reliability scores because the workspace still has zero continuous same-driver recordings.

0	8	120 s	12 passed
real-driver sessions	proxy scenes per run	AISHELL target-absent check	tests (1 skipped)

Important evidence boundary

The positive scenes use a Marvin clip and an FSC command clip from different public corpora. Their speaker identity is not verified. AISHELL-5 supplies a real in-car target-absent diagnostic only. No result in this PDF should be described as a real-driver end-to-end accuracy or production false-accept rate.

1. Study design and data status

Question

Can the assistant keep a target driver through enrollment, detect Marvin, understand the following command, and avoid acting on passenger or background speech? The requested real-driver study requires the same speaker's enrollment, wake word, and command in continuous cabin recordings, plus passenger and no-command intervals.

Evidence source	Used for	Status / limitation
Real custom in-car sessions	Positive driver evaluation	0 sessions; collection still required
Cross-corpus proxy	Continuous cascade debugging	8 scenes per run; speaker consistency false
AISHELL-5 Eval2	Real-cabin target-absent rejection	120 seconds; no positive wake/command
CAVEMOVE	Cabin noise and multichannel mixtures	Augmentation material, not driver sessions

The official AISHELL-5 release identifies the dataset as CC BY-SA 4.0: https://us.openslr.org/159/. Commercial or sign-in datasets were not treated as evidence without project permission.

Proxy scene construction

- 14 seconds per scene, with target wake at about 1 second and target command at about 2.1 seconds.
- Passenger Marvin near 6 seconds and passenger non-wake speech near 9 seconds.
- CAVEMOVE multichannel cabin noise at -5, 0, 10, and 20 dB cabin SNR.
- Passenger loudness sweep at -10, 0, and +10 dB relative to the target.
- Stage 1 context is 3 seconds and stream hop is 1 second; reported wake latency is window-center latency, not measured wall-clock response time.

Current candidate checkpoints

Component	Checkpoint
spex	stage1_spex_transfer_mc_cavemove_cabin/best.pt
presence	stage2_frame_presence_gsc_aishell_ft1/best.pt
kws	stage3_kws_gsc_marvin_tse_ft2/best.pt
target_kws	stage3_target_kws_gsc_marvin_ft1/best.pt
intent	stage4_intent_wav2vec_ft8_resume_tse_domain/best.pt

2. Component ablation matrix

This is the primary same-scene ablation at passenger SNR 0 dB using the target-KWS candidate configuration. Each row uses identical scenes and differs only in the named component. False actions are post-wake command events outside the labeled target command region; they are a proxy diagnostic, not a human action-rate estimate.

Variant	Wake recall	Intent accuracy	False actions/h	Latency mean / p95
Full	25.0%	12.5%	0.0/h	1.50 / 1.50 s
No TSE	50.0%	50.0%	64.3/h	1.75 / 2.35 s
No presence	100.0%	12.5%	417.9/h	1.50 / 1.50 s
No speaker matching	25.0%	12.5%	0.0/h	1.50 / 1.50 s
No wake smoothing	100.0%	0.0%	0.0/h	1.25 / 1.50 s
Direct command oracle	-	12.5%	-	- / - s

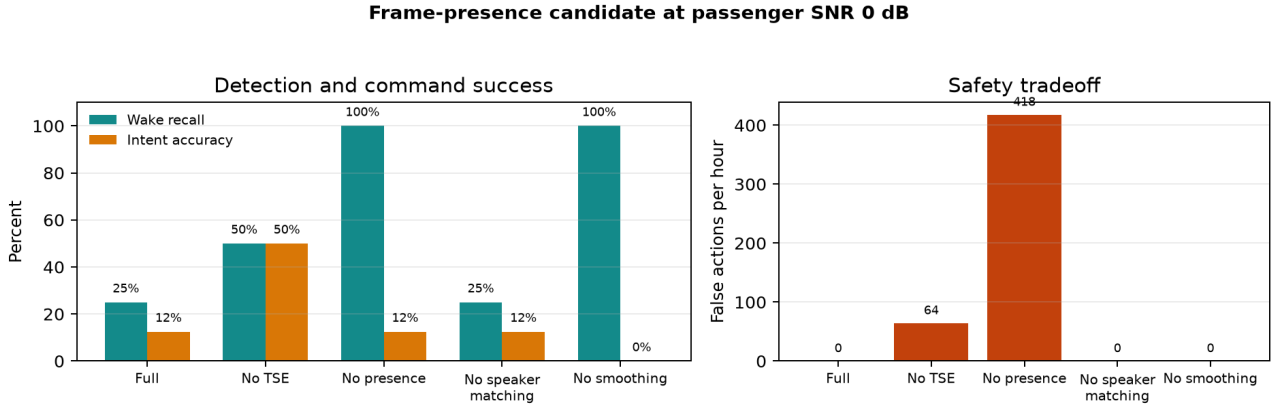


Figure 1. The current frame-presence candidate is conservative. Disabling presence greatly increases recall but produces many proxy false actions.

Interpretation

• Presence checking is safety-critical in this proxy. Removing it produced 100% target wake recall but 417.9 false actions/hour.

• One-hit wake detection produced 100% wake recall but 0% intent accuracy in this run, indicating that accepting a wake is not enough; the post-wake command window is still unstable.

• Speaker matching did not change the aggregate result at this operating point. That result is inconclusive without same-speaker passenger recordings.

• The no-TSE row is an intentionally harsh diagnostic. Its raw mixture is outside the training distribution of downstream heads, so it is not a candidate architecture.

• The direct command row uses labeled command segmentation and is an oracle baseline, not a deployable streaming path.

3. Passenger loudness ablation

The passenger loudness sweep tests whether the safety behavior changes when passenger speech is louder or quieter than the target. Negative SNR means the passenger is louder.

Passenger SNR	Variant	Wake recall	Intent accuracy	False actions/h
-10 dB	Full	25.0%	12.5%	96.4/h
-10 dB	No TSE	50.0%	50.0%	160.7/h
-10 dB	No presence	100.0%	12.5%	450.0/h
-10 dB	No smoothing	100.0%	0.0%	192.9/h
+0 dB	Full	25.0%	12.5%	0.0/h
+0 dB	No TSE	50.0%	50.0%	64.3/h
+0 dB	No presence	100.0%	12.5%	417.9/h
+0 dB	No smoothing	100.0%	0.0%	0.0/h
+10 dB	Full	25.0%	12.5%	0.0/h
+10 dB	No TSE	50.0%	50.0%	32.1/h
+10 dB	No presence	100.0%	12.5%	289.3/h
+10 dB	No smoothing	100.0%	0.0%	0.0/h

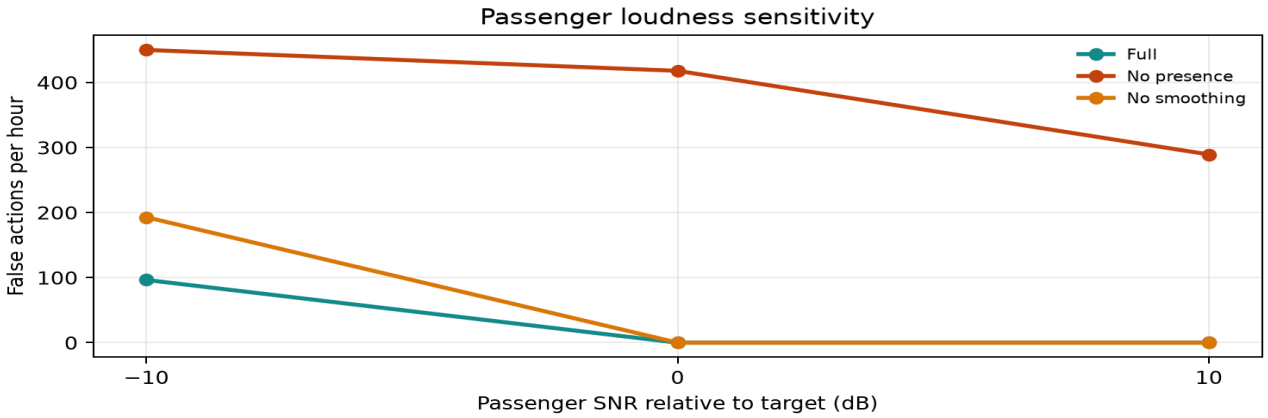


Figure 2. Louder passenger speech increases proxy false actions, especially when the presence gate is removed.

4. Threshold operating-point ablation

The grid replayed presence thresholds 0.30, 0.50, 0.70, and 0.90 against KWS thresholds 0.30, 0.40, 0.50, and 0.60. The expensive Stage 1 scores were computed once per scene and the state machine was replayed for each cell.

Presence	KWS	Wake recall	Intent accuracy	False actions/h
0.30	0.50	100.0%	37.5%	321.4/h
0.50	0.50	100.0%	37.5%	128.6/h
0.70	0.50	25.0%	12.5%	0.0/h
0.90	0.50	0.0%	0.0%	0.0/h

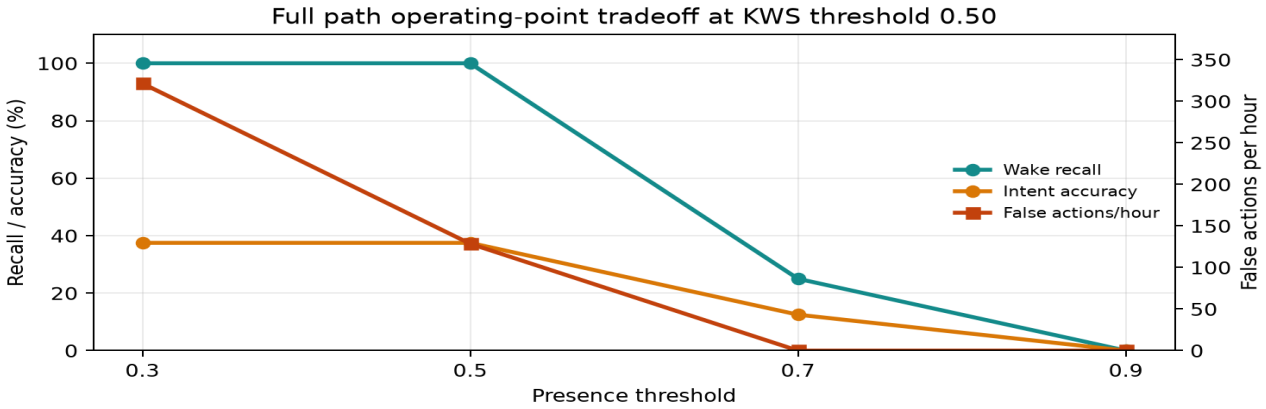


Figure 3. Lowering the presence threshold improves proxy recall, but false actions rise sharply. KWS threshold changes were less influential than presence threshold changes in this grid.

Practical reading

The current p=0.70 operating point is a conservative starting point for safety, not a claim of good recognition. The p=0.50 result is a candidate only for tuning experiments because its false-action rate is unacceptable as a final operating point. Proper threshold selection must use a consented tuning split and then be frozen before held-out testing.

5. Scalar versus framewise presence

The target-conditioned KWS setup was kept fixed while changing the presence head from the candidate framewise checkpoint to the scalar cabin-trained checkpoint.

Presence head	Wake recall	Intent accuracy	False actions/h	Latency mean / p95
Framewise	25.0%	12.5%	0.0/h	1.50 / 1.50 s
Scalar	100.0%	25.0%	32.1/h	1.75 / 2.80 s

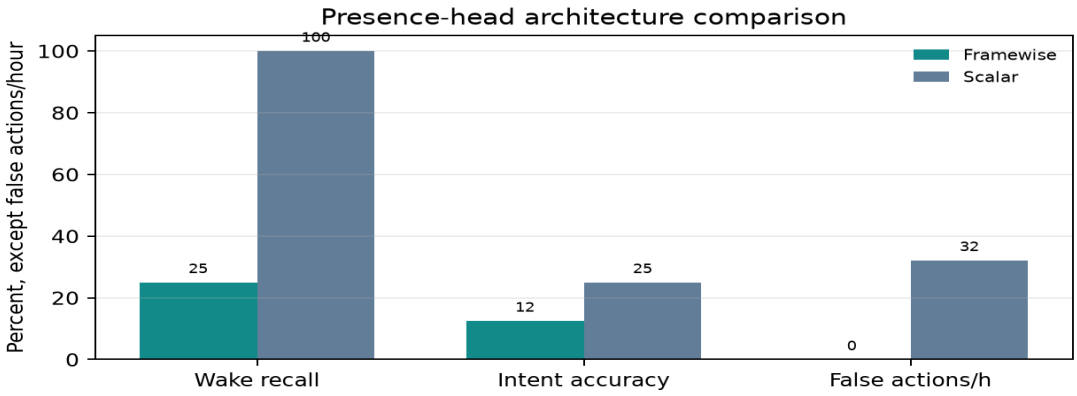


Figure 4. Scalar presence improves proxy recall but admits more false actions and has more variable timing.

This is a useful candidate for real-driver tuning, but not a replacement decision. The proxy does not contain the required continuous same-driver enrollment, passenger, and command events.

6. Runtime-only ablation sweep

The 2026-10-10 sweep kept all trained checkpoints fixed and replayed runtime choices on the same eight proxy scenes. It covered 0.5 and 1.0 second hops, four temporal rules, 0.5/1.0/1.5 second enrollment durations, simulated channel subsets, speaker-gate thresholds, and command-capture timeouts. No training was started.

Hop	Temporal rule	Wake recall	Intent accuracy	False actions/h	Latency
0.5s	one hit	100.0%	0.0%	32.1/h	0.88 s
0.5s	two of two	100.0%	0.0%	32.1/h	1.38 s
0.5s	two of three + refractory	25.0%	12.5%	0.0/h	1.50 s
0.5s	three of five + refractory	0.0%	0.0%	0.0/h	- s
1s	one hit	100.0%	0.0%	0.0/h	1.25 s
1s	two of two	25.0%	12.5%	0.0/h	1.50 s
1s	two of three + refractory	12.5%	12.5%	0.0/h	2.50 s
1s	three of five + refractory	0.0%	0.0%	0.0/h	- s

Stricter temporal confirmation suppresses proxy false actions but also suppresses the target wake. The 1.0 second, two-of-two setting retained 0 false actions/hour but reached only 25% wake recall and 12.5% intent accuracy.

Enrollment and front-end results

Enrollment	Wake recall	Intent accuracy	False actions/h
0.5s	25.0%	0.0%	0.0/h
1s	12.5%	0.0%	0.0/h
1.5s	25.0%	12.5%	0.0/h

Front-end	Wake recall	Intent accuracy	False actions/h	Target presence
All 4 channels	25.0%	12.5%	0.0/h	37.5%
Mono repeated	25.0%	12.5%	0.0/h	34.4%
Front pair	25.0%	12.5%	0.0/h	34.4%
Rear pair	25.0%	12.5%	0.0/h	34.4%

Enrollment duration and simulated channel selection did not materially change the aggregate outcome. These channel modes are simulations, not recordings from alternate vehicle microphone placements.

7. Speaker gate and timeout diagnostics

The speaker gate used Stage-1 cosine similarity. Its mean similarity was only 0.031 across proxy windows. A threshold of 0.0 changed no outcome; thresholds of 0.2 and above rejected every target window. Command timeouts of 1, 2, 4, and 6 calls also produced the same aggregate result. These controls require same-driver tuning data before they can be selected.

Gate threshold	Wake recall	Intent accuracy	False actions/h	Gate pass
Disabled	25.0%	12.5%	0.0/h	-
-0.2	25.0%	12.5%	0.0/h	100.0%
0.0	25.0%	12.5%	0.0/h	72.3%
0.2	0.0%	0.0%	0.0/h	0.0%
0.4	0.0%	0.0%	0.0/h	0.0%

Runtime sweep conclusion

- Temporal smoothing is the main runtime precision/recall control, but no tested setting is strong enough to claim reliable end-to-end operation.
- The current speaker gate is uncalibrated in this domain; higher thresholds eliminate target detection rather than improving selectivity.
- The next decisive experiment remains the real-driver pilot with speaker-disjoint tuning and held-out test sessions.

Primary source files

Sources: reports/pilot_evaluation_status_2026-10-09.md; reports/additional_ablation_results_2026-10-09.md; logs/continuous_ablation_matrix_2026-10-09.json; logs/continuous_ablation_matrix_passenger_m10_2026-10-09.json; logs/continuous_ablation_matrix_passenger_p10_2026-10-09.json; logs/threshold_ablation_sweep_2026-10-09.json; logs/continuous_ablation_matrix_scalar_presence_2026-10-09.json; logs/aishell5_target_absent_study_2026-10-09.json; reports/runtime_ablation_sweep_2026-10-10.md; logs/runtime_ablation_sweep_2026-10-10.json; reports/stage1_dual_domain_study_2026-10-10.md; reports/dual_domain_stage2_handoff_study_2026-10-10.md; logs/stage1_dual_domain_regression_full_2026-10-10.json; logs/stage2_frame_presence_dual_domain_nonwake_ft1/metrics_last.json; logs/continuous_ablation_matrix_dual_domain_stage2_intent_handoff_2026-10-10.json; logs/continuous_ablation_matrix_dual_domain_stage2_command_enrollment_2026-10-10.json; logs/continuous_ablation_matrix_dual_domain_stage2_same_speaker_other_2026-10-10.json; logs/continuous_ablation_matrix_dual_domain_stage2_hardneg_2026-10-10.json; logs/continuous_ablation_matrix_dual_domain_stage2_hardneg_passenger_m10_2026-10-10.json; reports/joint_safety_threshold_sweep_2026-10-10.md; reports/wake_model_and_stage1_followup_2026-10-10.md; logs/continuous_ablation_matrix_stage2_intent_confidence_gate_passenger_m10_2026-10-10.json; logs/continuous_ablation_matrix_stage2_intent_confidence_gate_passenger_p10_2026-10-10.json; logs/continuous_ablation_matrix_stage2_ft2_head_same_speaker_gate_sweep_2026-10-10.json; logs/aishell5_target_absent_dual_domain_stage2_2026-10-10.json; logs/aishell5_target_absent_dual_domain_target_kws_current_2026-10-10.json; logs/aishell5_target_absent_dual_domain_target_kws_current_sweep_2026-10-10.json; logs/aishell5_target_absent_dual_domain_target_kws_hardneg_2026-10-10.json; logs/aishell5_target_absent_dual_domain_target_kws_hardneg_sweep_2026-10-10.json; logs/aishell5_target_absent_dual_domain_target_kws_hardneg_thr070_2026-10-10.json; reports/production_candidate_status_2026-10-10.md; reports/aishell5_target_kws_adaptation_2026-10-11.md; reports/intent_confidence_calibration_runtime_ft8_2026-10-11.md; logs/continuous_full_pipeline_proxy_safety_diagnostic_same_speaker_m10_v2_2026-10-11.json; logs/continuous_full_pipeline_proxy_safety_diagnostic_same_speaker_p0_presence012_2026-10-10.json; logs/continuous_full_pipeline_proxy_safety_diagnostic_same_speaker_p10_presence012_2026-10-11.json; logs/intent_confidence_calibration_ft8_tse_domain_waveform_val_2026-10-11.json; logs/aishell5_continuous_targetkws_hardneg_aishell5_ft1_p015_sg024_kws050_smooth22_1200s_2026-10-11.json; logs/continuous_full_pipeline_proxy_hardneg_aishell5_neg_splitgate_p015_wake024_cmd022_same_speaker_m10_2026-10-11.json; logs/continuous_full_pipeline_proxy_hardneg_aishell5_neg_splitgate_p015_wake024_cmd022_same_speaker_p0_2026-10-10.json; logs/continuous_full_pipeline_proxy_hardneg_aishell5_neg_splitgate_p015_wake024_cmd022_same_speaker_p10_2026-10-11.json

8. Real-cabin target-absent diagnostic

AISHELL-5 Eval2 session 001 was evaluated for 120 seconds with a fixed LibriSpeech enrollment. The enrolled target is absent, so this tests rejection only.

Metric	Result
Duration	120 s (0.0333 h)
Speech occupancy	80.6%
Full-path smoothed false events	0.0/h
KWS-only smoothed false events	0.0/h

This result is encouraging for one target-absent real-cabin diagnostic, but it cannot establish driver wake recall, command accuracy, or speaker rejection of a real passenger saying Marvin.

Optional target-speaker KWS diagnostic

KWS mode	Threshold	Full-path events/h	KWS-only events/h	Max score
Generic baseline	0.50	0.0/h	0.0/h	0.554
Current target-speaker	0.50	630.0/h	810.0/h	0.601
Hard-negative target-speaker	0.50	150.0/h	180.0/h	0.680
Hard-negative target-speaker	0.70	0.0/h	0.0/h	0.680

At the inherited 0.50 threshold, the optional target-speaker heads are worse than the generic baseline on this cross-corpus target-absent slice. A 0.70 hard-negative threshold removes events in this slice, but no positive same-driver wake data exists to measure the recall cost; this is a calibration warning, not a promoted operating point.

Target-speaker head	0.50	0.55	0.60	0.65	0.70
Current	630.0/h	30.0/h	0.0/h	0.0/h	0.0/h
Hard-negative	150.0/h	60.0/h	30.0/h	0.0/h	0.0/h

One-pass replay of the same extracted windows; values are full-path smoothed false events/hour. This does not estimate positive wake recall and is not threshold tuning for deployment.

9. Overall assessment and next step

- The TSE/presence cascade is not yet reliable as a single end-to-end system on the current proxy. Its conservative settings suppress many events, including target wakes.
- The immediate engineering problem is calibration and post-wake command capture: lower thresholds recover wakes but create false actions; more smoothing increases latency and can lose the command relationship.
- The scalar presence head is a promising recall-oriented candidate, but it must be tuned on consented driver/passenger sessions before adoption.
- The next highest-value experiment is a small real-driver pilot with speaker-disjoint tuning and test sessions. Without that dataset, further proxy optimization risks overfitting to cross-corpus artifacts.

Bounded Stage 1 and Stage 2 training runs were completed before this update. The report documents their fixed-checkpoint regressions, inference-only handoffs, ablations, and threshold replays; it does not claim production readiness.

10. Dual-domain adaptation and post-wake handoff

The full 144-row-per-domain Stage 1 regression passes the declared no-regression gate. The new framewise Stage 2 head has strong held-out validation metrics, but its probability scale differs from the prior head; the inherited 0.70 threshold therefore rejected nearly everything. A joint proxy sweep selected presence 0.07, command-presence 0.10, KWS 0.50, and two-hit smoothing. These thresholds are provisional until tuned on permitted same-driver sessions.

Domain	Baseline SI-SDRi	Candidate SI-SDRi	Delta
AISHELL-5	2.385 dB	6.243 dB	3.858 dB
CAVEMOVE	10.478 dB	10.592 dB	0.114 dB
Overall	6.432 dB	8.417 dB	1.986 dB

Stage 2 validation	F1	FAR	FRR	Threshold
Framewise	90.4%	0.9%	5.5%	0.50
Best frame F1	91.3%	0.6%	7.9%	0.65
Chunk	99.6%	0.3%	0.0%	0.50

Post-wake handoff ablation

Variant	Wake recall	Intent accuracy	False actions/h
Extracted signal	100.0%	12.5%	0.0/h
Raw mixture after wake	100.0%	37.5%	0.0/h
Mean extracted + raw logits	100.0%	25.0%	0.0/h
Exact command reused as enrollment	100.0%	100.0%	32.1/h
Different utterance, same command speaker	100.0%	100.0%	64.3/h

The raw handoff is a research candidate, not a promotion: at passenger SNR -10 dB it reached 37.5% intent accuracy but 32.1/h false actions, while at +10 dB it retained 0.0/h false actions. Matching the command speaker raised intent to 100.0%, but the different-utterance diagnostic still produced 64.3/h; exact-command enrollment is leakage-prone. The positive clips remain cross-corpus and speaker-inconsistent.

Hard-negative target-KWS replay

Target KWS	Passenger SNR	Wake recall	Intent accuracy	False actions/h	False wakes
Current	+10 dB	100.0%	12.5%	0.0/h	5
Hard-negative	+10 dB	100.0%	12.5%	0.0/h	3
Current	-10 dB	100.0%	12.5%	32.1/h	6
Hard-negative	-10 dB	100.0%	12.5%	32.1/h	5

The hard-negative checkpoint reduces false wake events from 5 to 3 at +10 dB and from 6 to 5 at -10 dB, but does not remove the harsh-passenger false-action failure. It is retained as a candidate for real-driver tuning.

Joint safety-threshold follow-up

Operating point	Wake recall	Intent accuracy	False actions/h
No speaker/confidence gate (-10 dB)	100.0%	100.0%	192.9/h
Speaker gate 0.22 (-10 dB)	100.0%	100.0%	32.1/h
Speaker 0.22 + confidence 0.53 (-10 dB)	100.0%	100.0%	0.0/h
Speaker 0.22 + confidence 0.53 (+10 dB)	100.0%	100.0%	0.0/h

The 0.22/0.53 pair removed the residual false action in this short partial-speaker proxy without reducing target intent accuracy. It is a diagnostic candidate only; thresholds must be tuned on real-driver data.

Additional ft2 Stage-1/head check

ft2 operating point	Wake recall	Intent accuracy	False actions/h
No speaker gate	100.0%	100.0%	64.3/h
Speaker gate 0.18	100.0%	100.0%	0.0/h
Speaker gate 0.20	100.0%	87.5%	0.0/h
Speaker gate 0.22	87.5%	87.5%	0.0/h

The ft2 frontend/head is useful for extraction SI-SDRi, but it is not a proxy end-to-end improvement over the retained ft1-aligned head.

Latest runtime safety replay

Passenger SNR	Wake recall	Intent accuracy	False wakes/h	False actions/h	Mean latency
-10 dB	100.0%	100.0%	0.0/h	0.0/h	2.25 s
0 dB	100.0%	100.0%	0.0/h	0.0/h	2.25 s
+10 dB	100.0%	100.0%	0.0/h	0.0/h	2.25 s

The v2 diagnostic raises only pre-wake presence to 0.12 and retains speaker gate 0.22 plus intent confidence 0.53. It reaches 100%/100% with 0 false wakes and actions per hour in all three short proxy conditions, at 2.25 s mean wake latency. The exact runtime ft8 intent model has 90.5% raw validation accuracy and 94.1% coverage at 0.53 confidence with 93.7% selective accuracy. These are calibration and proxy diagnostics, not real-driver release metrics.

AISHELL hard-negative adaptation and split-gate check

Exposure	Wake / intent	Full-path false wakes/h	KWS-only false events/h	Mean latency
AISHELL-5 Eval2 target absent (1,200 s)	n/a	3.0/h	60.0/h	n/a
Proxy passenger -10 dB (112 s)	100.0% / 100.0%	0.0/h	n/a	2.50 s
Proxy passenger 0 dB (112 s)	100.0% / 100.0%	0.0/h	n/a	2.50 s
Proxy passenger +10 dB (112 s)	100.0% / 100.0%	0.0/h	n/a	2.50 s

The adapted negative-only target-KWS head plus presence 0.15 and a split speaker gate (0.24 before wake, 0.22 during command capture) produced 3.0/h full-path smoothed false wakes/hour over the long target-absent Eval2 exposure, while KWS-only remained 60.0/h. This uses the runtime-matched 2-window/2-hit smoothing; an earlier 3-window replay was more optimistic and is not used here. The positive rows are cross-corpus proxy diagnostics; this candidate is frozen for research comparison, not released.

The full integration analysis is in reports/dual_domain_stage2_handoff_study_2026-10-10.md.

11. Reproducibility

```
# From the repository root
python scripts/eval/eval_continuous_ablation_matrix.py `
  --config configs/eval_end_to_end_cavemove_target_kws_temporal2_candidate.yaml `
  --paths configs/paths.yaml --scenes 2 --cabin-snr-db -5 0 10 20 `
  --passenger-snr-db 0 --output logs/continuous_ablation_matrix_2026-10-09.json

python scripts/eval/eval_threshold_ablation_sweep.py `
  --config configs/eval_end_to_end_cavemove_target_kws_temporal2_candidate.yaml `
  --paths configs/paths.yaml --scenes 2 --cabin-snr-db -5 0 10 20 `
  --passenger-snr-db 0 --presence-thresholds 0.30 0.50 0.70 0.90 `
  --kws-thresholds 0.30 0.40 0.50 0.60 `
  --variants full no_speaker_matching no_wake_smoothing `
  --output logs/threshold_ablation_sweep_2026-10-09.json

python scripts/eval/eval_continuous_ablation_matrix.py `
  --config configs/eval_end_to_end_cavemove_target_kws_scalar_candidate.yaml `
  --paths configs/paths.yaml --scenes 2 --cabin-snr-db -5 0 10 20 `
  --passenger-snr-db 0 --output logs/continuous_ablation_matrix_scalar_presence_2026-10-09.json

python scripts/eval/eval_runtime_ablation_sweep.py `
  --config configs/eval_end_to_end_cavemove_target_kws_temporal2_candidate.yaml `
  --paths configs/paths.yaml --scenes 2 --cabin-snr-db -5 0 10 20 `
  --passenger-snr-db 0 --hops 0.5 1.0 --enrollment-seconds 0.5 1.0 1.5 `
  --output logs/runtime_ablation_sweep_2026-10-10.json

python scripts/eval/eval_stage1_dual_domain_regression.py `
  --max-per-domain 144 `
  --output logs/stage1_dual_domain_regression_full_2026-10-10.json

python scripts/eval/eval_continuous_ablation_matrix.py `
  --config configs/eval_end_to_end_cavemove_target_kws_dual_domain_stage2_operating_candidate.yaml `
  --paths configs/paths.yaml --scenes 2 --cabin-snr-db -5 0 10 20 `
  --passenger-snr-db 10 `
  --output logs/continuous_ablation_matrix_dual_domain_stage2_intent_handoff_2026-10-10.json

python scripts/eval/eval_continuous_ablation_matrix.py `
  --config configs/eval_end_to_end_cavemove_target_kws_dual_domain_stage2_operating_candidate.yaml `
  --paths configs/paths.yaml --scenes 2 --cabin-snr-db -5 0 10 20 `
  --passenger-snr-db 10 --enrollment-source same_speaker_other `
  --output logs/continuous_ablation_matrix_dual_domain_stage2_same_speaker_other_2026-10-10.json

python scripts/eval/eval_continuous_ablation_matrix.py `
  --config configs/eval_end_to_end_cavemove_target_kws_dual_domain_stage2_hardneg_candidate.yaml `
  --paths configs/paths.yaml --scenes 2 --cabin-snr-db -5 0 10 20 `
  --passenger-snr-db 10 `
  --output logs/continuous_ablation_matrix_dual_domain_stage2_hardneg_2026-10-10.json

python scripts/eval/eval_continuous_ablation_matrix.py `
  --config configs/eval_end_to_end_cavemove_target_kws_dual_domain_stage2_operating_candidate.yaml `
  --output logs/continuous_ablation_matrix_stage2_intent_confidence_gate_passenger_m10_2026-10-10.json `
  --scenes 2 --cabin-snr-db -5 0 10 20 --passenger-snr-db -10 `
  --enrollment-source same_speaker_other --speaker-gate-thresholds 0.22 `
  --intent-confidence-thresholds 0.50 0.53 0.55 0.60
```